UCD Michael Smurfit Graduate Business School

Does Music Sentiments affect US Stock Market Returns?

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I hereby declare that this project report is the result of our own work, prepared independently without any unauthorized assistance, and has not been copied or plagiarized from any other source.

Abstract

In this research, the role of music sentiment, captured by the daily valence scores of Spotify, as a predictor of U.S. stock returns is explored. Using daily data for the S&P 500 index for the year 2023, both simultaneous and lagged tests of the relationship between sentiment and stock performance are conducted. Ordinary least squares regressions are compared with autoregressions for return persistence, whereas robust standard error tests including heteroskedastic-robust (HC1) standard errors and Newey–West HAC models are carried out for robust inference, with exhaustive testing for model validity using ARIMA tests of residuals.

Empirical evidence shows that there is no significant relationship between musical sentiment and daily returns of the S&P 500. Coefficient sizes are consistently small and insignificant for all models, including lagged models, autoregressive models, and confidence intervals constructed via bootstrapping. Tests of convergence confirm that the models meet the assumptions of regression analysis and that residuals are white noise.

These results are in contrast with previous international evidence and indicate that the influence of sentiment effects via music consumption on returns could be insignificant in large, liquid, and information-efficient markets like the U.S. markets. These findings add to the literature of behavioral finance by indicating that return predictability via sentiment is market specific.

1 Introduction

1.1 Motivation

From Long time it has been believed that economic variables like earnings, interest rates, GDP etc. are the main factors influencing financial markets. Research on behavioral finance focuses on how important the mood and sentiments of investor are in shaping market results. It has been observed that music can act as a universal medium for expressing emotions. Daily fluctuations in music sentiment, which are measured by characteristics like "valence" on streaming services like Spotify, may reflect broader societal emotions that affect investors' trading behavior and risk tolerance.

We can obtain a fresh understanding of how non-financial, cultural factors may impact financial markets by investigating the possible connection between musical sentiment and stock market returns. By offering empirical proof of how social mood affects market dynamics, this study also adds to the expanding body of research on behavioral finance.

1.2 Behavioral Finance Context

According to conventional financial theory, investors behave rationally and base their choices only on the information about fundamentals that is currently available. However, behavioral finance recognizes that systematic departures from rationality can result from psychological, emotional, and social influences. Decision-making in the face of uncertainty, risk perception, and cognitive processing are all impacted by emotional states. Prior research has connected mood to anomalies in asset pricing, volatility, and trading intensity.

A distinct and quantifiable method of capturing the everyday collective mood is through musical sentiment. While negative sentiment may be linked with pessimistic and more cautious behavior in investment, positive sentiment may display increased optimistic behavior and an ability for taking risks. Studying this relationship can help us understand how social and cultural signals affect market results.

1.3 Overview of Edmans et al. Paper

Edmans et al.'s paper "Music sentiment and stock returns around the world" investigates whether popular music's emotional content is correlated with stock returns in various nations. The authors examine the same-day and lagged effects of music sentiment on market returns using Spotify streaming data and matching national stock indices. They consider whether shifts in valence, which represent either positive or negative mood swings, can forecast market performance.

The study supports the idea that social mood, as reflected in cultural consumption,

can have quantifiable effects on financial markets by finding a systematic relationship between music sentiment and subsequent stock returns. The paper presents an innovative method for researching sentiment-driven trading and behavioural finance by connecting music and market returns.

1.4 Replication Goals and Contribution

The main goal of this project is to replicate the method used by Edmans et al. We will analyze daily Spotify sentiment and S&P 500 returns for 2023. Specifically, this project aims to:

- **Data Collection and Preprocessing:** Gather daily music sentiment data (valence) from Spotify and daily S&P 500 stock returns. We will align both datasets to business-day frequency and address any missing values.
- **OLS Estimation:** Run ordinary least squares regressions to examine same-day and lagged connections between music sentiment and stock returns. I also included lagged dependent variables to check serial correlation in returns.
- **Model Validation and Diagnostics:** This step acts as a traditional OLS assumptions by testing residuals for normality, heteroskedasticity, and autocorrelation. Use robust standard errors (HC1) and Newey–West HAC estimates to fix any potential issues.
- **Residual Analysis with ARIMA Models:** Explore autocorrelation patterns in residuals using ARIMA modeling to find any time-related dependencies not captured by the OLS method.
- **Interpretation and Insights:** Evaluate the strength and importance of any observed relationships, compare them with Edmans et al.'s findings, and discuss possible economic and behavioral impacts.

By replicating the study, this project confirms the original findings, checks the strength of OLS results in a daily data environment, and investigates extra insights such as the impact of lagged sentiment effects. The study helps us understand how cultural and emotional signals, like music, can affect financial markets. This adds to traditional economic analyses.

2 Hypothesis

This study aims to investigate the potential predictive power of non-financial indicators of societal mood, such as music sentiment, for stock market returns. In particular, the analysis looks into the connection between daily sentiment on Spotify music and the S&P 500 index's daily returns in 2023.

The main Research Question:

Does Spotify valence, a measure of daily music sentiment, predict daily returns on the S&P 500 stock market?

The following theories are examined in order to formally evaluate this relationship:

Null Hypothesis (H_0)

S&P 500 stock market returns are unaffected by daily music sentiment.

Alternative Hypothesis (H_1)

Future S&P 500 stock market returns are predicted by higher music sentiment (higher Spotify valence).

Ordinary least squares (OLS) regression models with contemporaneous and lagged sentiment measures are used to assess these hypotheses, along with robustness tests that take autocorrelation, heteroskedasticity, and time-series dynamics into account.

3 Literature Review

3.1 Sentiment and Financial Markets

The relation between sentiment and asset prices has become the most well-debated issue of behavioral finance. Traditional finance theory premises prices must reflect all available information, while investors are usually rational. However, empirical evidence has brought forth a challenge to this assumption, indicating that investors' emotions, mood, and psychological biases can affect trading behavior, volatility, and asset pricing.

Many different proxies have been considered: investor surveys, news articles, social media activity, and online search trends. For example, Baker and Wurgler (2006) showed that investor sentiment explains cross-sectional variation in stock returns, and particularly, that these effects are strongest for high-volatility or difficult-to-arbitrage stocks. Similarly, Tetlock (2007) considered financial text and found that negative news media sentiment reduces next-day stock prices, thus showing how emotions encoded in text can impact market behavior.

3.2 Music as a Sentiment Proxy

People now treat music listening as a reliable gauge of public mood. Every track yields numbers for positivity, energy and tempo. Spotify and similar platforms release daily streaming figures that mirror how a nation feels.
Edmans et al. (2023) were the first to apply this measure to finance — they linked each country's daily Spotify positivity score to its main stock index plus found that a rise in positivity predicts higher returns the next day, while a drop forecasts lower returns. The result adds a cultural layer to behavioral finance — investor mood is not limited to financial headlines — it also reflects wider societal emotion.

3.3 Time Series Analysis and OLS Regression

OLS regression is still a key method for looking at how things relate in time series data. When it comes to financial returns, OLS can show how things like past sentiment impact what happens in the market. Thing is, time series data often has autocorrelation, heteroskedasticity, and non-stationarity, which messes with the standard OLS assumptions. To deal with these issues, people try things like:

- **Lagged variables:** Adding past sentiment or returns to factor in how things change over time.
- **Corrected standard errors (HC1):** Fixing heteroskedasticity to get results you can trust.

- **Newey–West HAC standard errors:** used for both autocorrelation and heteroskedasticity in the remaining values.

These methods confirm that coefficient estimates remain unbiased throughout and that hypothesis tests are relevant despite observed that there are few potential violations of OLS assumptions.

3.4 Residual Analysis and ARIMA Models

In time series analysis, residual diagnostics play a vital role in ensuring the adequacy of a model. Despite accounting for lagged predictors, residuals may still exhibit temporal structure, suggesting that additional dynamics remain unmodeled. To address this, ARIMA models are often employed to capture autocorrelation and forecast residual behavior. By incorporating ARIMA analysis, researchers can identify overlooked patterns in residuals, ultimately leading to improved model specification and more accurate forecasting.

3.5 Gaps and Extensions in Current Project

While Edmans et al. (2020) focus on multi-country weekly data, this project extends their analysis to daily frequency for a single major market (S&P 500). By increasing temporal granularity:

- We can explore whether same-day and lagged music sentiment effects persist daily.
- The project provides an opportunity to test the robustness of OLS and HAC corrections in high-frequency settings.
- Residual ARIMA analysis is applied to detect finer-scale autocorrelation patterns not captured by weekly models.

4 Data and Methods

4.1 Data Sources and Collection

By combining two sets of information—financial market and music mood—this study aims to determine whether there are any correlations between how people feel about their songs and how well they do financially. To do this, we relied on two key sources:

4.1.1 Spotify Sentiment Data

The Spotify data provided us with daily metrics for each song. For each song we can see:

- track_id, track_title, artist
- number of streams (the number of times the song was played on Spotify in one day)
- valence (measuring how happy/positive the song is on a scale from 0 to 1)
- energy (how energetic and stimulating the song is)
- tempo (with BPM as a standard measure of speed)

The data includes the year 2023 and also information from more than 18,000 songs in various different countries. Data is collected from 01/01/2023- 31/12/2023

4.1.2 Stock Market Data

Using the yfinance Python library, we extracted daily closing prices of the S&P 500 index.
This data spans the same time period as the Spotify dataset, which allows for temporal alignment.
We have created CSV files for both datasets so they can be easily reproduced: spotify_sentiment.c and sp500_logret.csv.

4.2 Data Preprocessing

4.2.1 Music Sentiment Aggregation

To construct daily measures of collective mood, track-level data were aggregated by date. Similarly, avg_energy was computed as a stream-weighted daily average.
This aggregation captures the overall emotional tone of music listened to by users on each day.

4.2.2 Financial Returns

Log returns of the S&P 500 were computed as:

$$\text{SP500 logret}_t = \ln(P_t) - \ln(P_{t-1})$$

This transformation ensures that returns are continuously compounded and approximately normally distributed.

4.2.3 Business Day Alignment

Both datasets were converted to business day frequency (B in pandas).
Missing values of prices , such as weekends or holidays in either dataset, were forward filled (ffill) to maintain alignment.
The final merged dataset contains 258 observations (business days) with no missing data for either variable.

4.3 Variable Description

Table 1: Variable Descriptions

Variable	Description	Units / Range
SP500_logret	Daily log return of S&P 500 index	Continuous, [-0.02, 0.02]
Avg_Valence	Daily average valence weighted by streams	0–1
Avg_Energy	Daily average energy weighted by streams	0–1
Valence_lag1	Previous day's avg_valence	0–1
SP500_logret_lag1	Previous day's S&P 500 log return	Continuous

Interpretation:

Descriptive statistics show that avg_valence is relatively stable, fluctuating between 0.36 and 0.60 with a mean of 0.50. The S&P 500 log returns are near zero on average (0.086%) with a standard deviation of 0.81%, reflecting typical daily volatility.

4.4 Descriptive Statistics and Visualizations

4.4.1 Summary Table

Table 2: Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
SP500_logret	258	0.000857	0.0081	-0.0202	-0.0039	0.0005	0.0061	0.0226
Avg_Valence	258	0.4988	0.0399	0.3587	0.4743	0.5006	0.5285	0.5987

Interpretation:
The mean and standard deviation values exhibits that music sentiment measure by Spotify Valence is stable over time. The

4.4.2 Time Series Plots

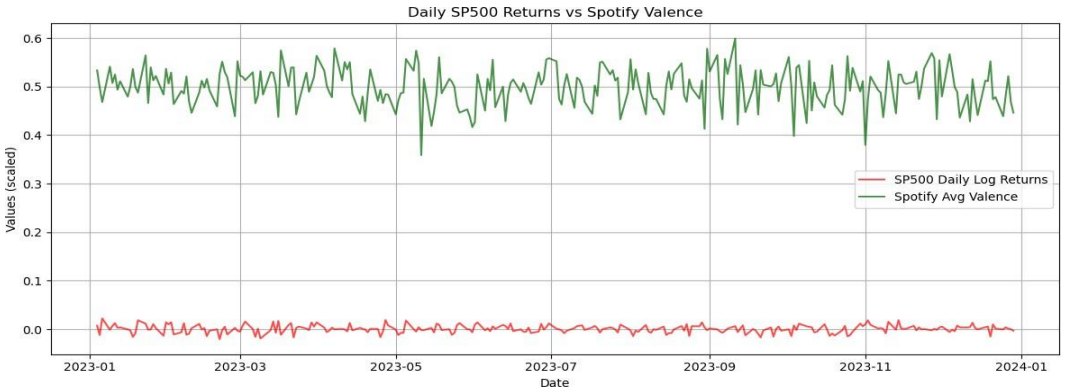


Figure 1: Daily S&P 500 Log Returns and Spotify Valence (2023)

4.4.3 Distribution Plots

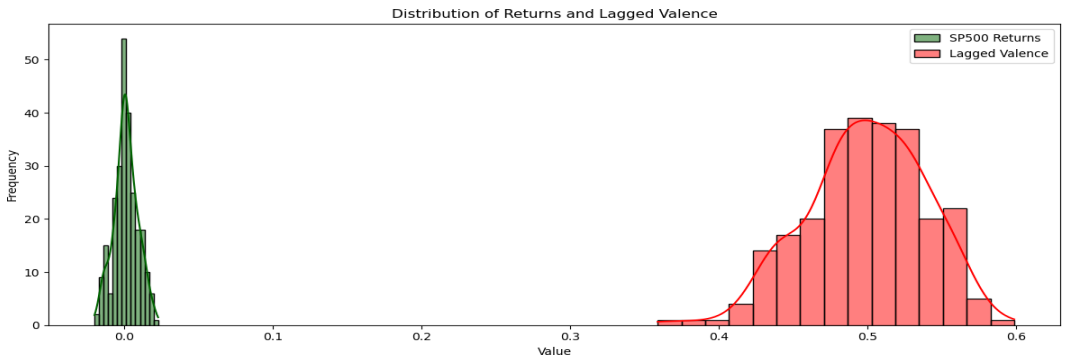


Figure 2: Distribution of Daily Returns and Music Valence

5 Empirical Results

5.1 Overview

This section will introduce data driven observations on results obtained from regression equation analysis, diagnostic testing, and ARIMA procedures. The focus will be on investigating if daily Spotify music valence influences daily U.S. stock market returns in 2023. We will attempt various OLS equation analyses, analyze violations on classical assumptions, and explore the characteristics of residuals via ARIMA.

The analysis reveals no statistically significant relationship between music sentiment and equity returns on U.S. stocks. But there are some aspects revealed from diagnostics and modeling performed on time series.

5.2 OLS Regression Results

We first estimate a simple OLS model:

$$\text{SP500_logret}_t = \alpha + \beta \cdot \text{Avg_Valence}_t + \varepsilon_t$$

Table 3: Same-Day OLS Regression Results

Coefficient	Value
β (Valence)	-0.0061
p-value	0.595
R^2	0.001

```
OLS (same-day) summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:           SP500_logret   R-squared:                       0.001
Model:                            OLS   Adj. R-squared:                 -0.003
Method:                 Least Squares   F-statistic:                    0.2826
Date:                Wed, 10 Dec 2025   Prob (F-statistic):              0.595
Time:                        22:57:06   Log-Likelihood:                 876.78
No. Observations:                 258   AIC:                            -1750.
Df Residuals:                     256   BIC:                            -1742.
Df Model:                           1
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0039      0.006      0.687      0.492      -0.007       0.015
avg_valence   -0.0061      0.011     -0.532      0.595      -0.029       0.016
==============================================================================
Omnibus:                        0.055   Durbin-Watson:                   1.943
Prob(Omnibus):                  0.973   Jarque-Bera (JB):                0.162
Skew:                          -0.014   Prob(JB):                        0.922
Kurtosis:                       2.881   Cond. No.                         31.3
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

Interpretation:

There is no evidence that same-day aggregate music sentiment has any explanatory power for daily stock returns. The avg_valence coefficient (-0.0061) is small, negative, and statistically insignificant, indicating no relationship between music sentiment and stock market. The low R^2 value further suggests that music sentiment explains virtually none of the variation in daily returns.

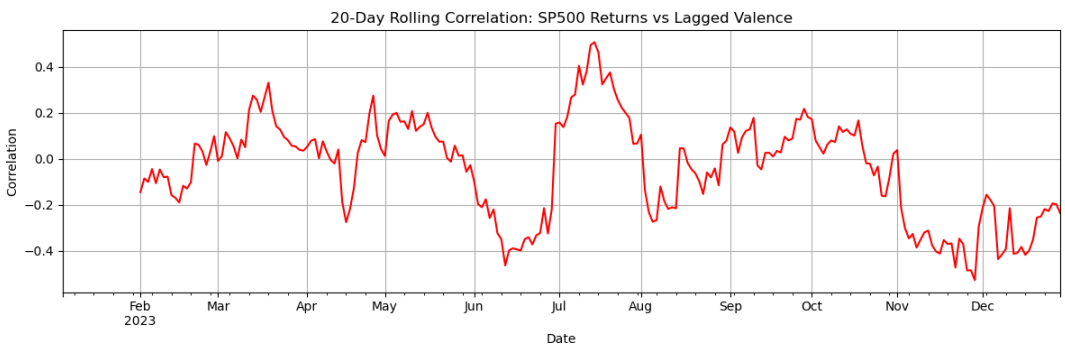


Figure 3: 20-Day Rolling Correlation

Interpretation:

A rolling 20-day correlation between lagged valence and S&P 500 returns was also computed to capture the time-varying nature of the relationship. This time-varying analysis complements the static OLS results by showing that the sentiment–return relationship remains weak throughout the sample.

5.3 Lagged Effect: Does Yesterday's Mood Predict Today's Market?

To test causality, a one-day lag model is estimated:

$$\text{SP500 logret}_t = a + \beta \cdot \text{Valence}_{t-1} + \varepsilon_t$$

Table 4: Lagged Valence OLS Regression (HC1 Robust SE)

	Value
β	-0.0093
p-value	0.446
R^2	0.003
N	257

```
OLS (lag-1) summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:           SP500_logret   R-squared:                       0.002
Model:                            OLS   Adj. R-squared:                 -0.002
Method:                 Least Squares   F-statistic:                    0.5820
Date:                Wed, 10 Dec 2025   Prob (F-statistic):              0.446
Time:                        22:57:06   Log-Likelihood:                 873.38
No. Observations:                 257   AIC:                            -1743.
Df Residuals:                     255   BIC:                            -1736.
Df Model:                           1
Covariance Type:                  HC1
==================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
const             0.0055      0.006      0.887      0.375      -0.007       0.018
Valence_lag1     -0.0093      0.012     -0.763      0.446      -0.033       0.015
==============================================================================
Omnibus:                        0.114   Durbin-Watson:                   1.932
Prob(Omnibus):                  0.945   Jarque-Bera (JB):                0.225
Skew:                          -0.035   Prob(JB):                        0.893
Kurtosis:                       2.873   Cond. No.                         31.4
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

Interpretation

This table evaluates whether music sentiment has predictive power for next – day market returns. The coefficient on lagged variable (-0.0093) remains statistically insignificant even after correcting for heteroskedasticity using HCI robust Standard errors.

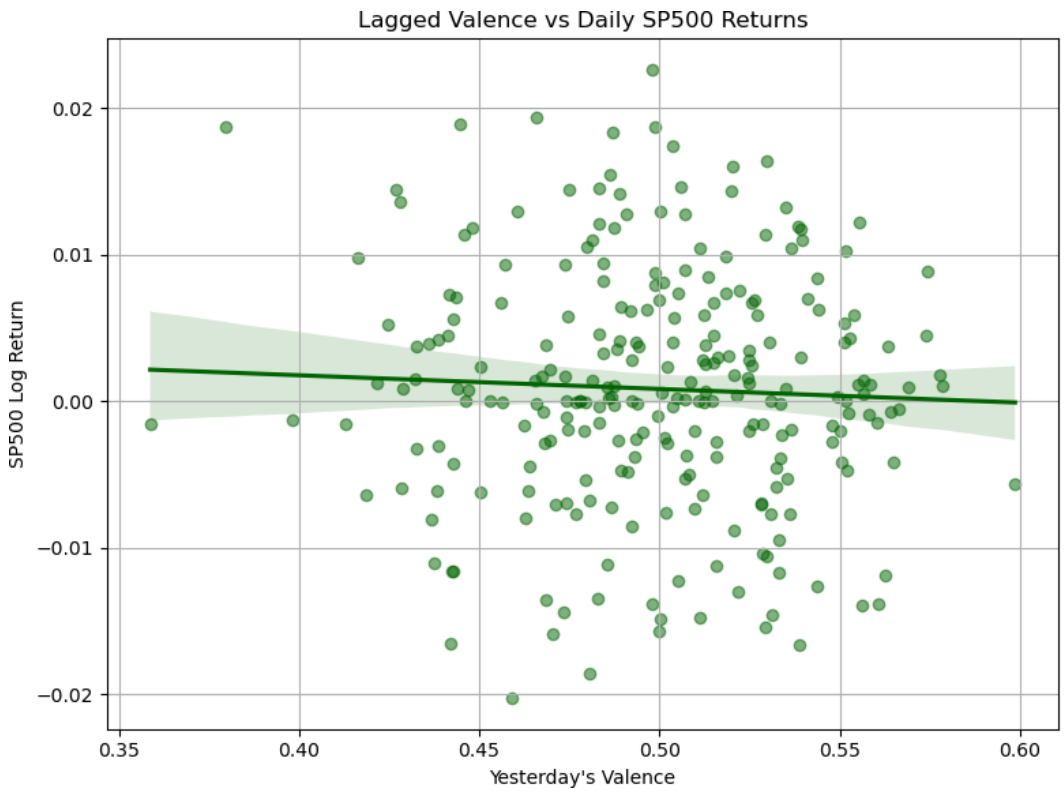


Figure 4: Lagged Valence vs Daily S&P 500 Returns

Interpretation:

The graphs shows the relationship between lagged valence and daily S&P 500 returns, this scatter plot with the regression line gives us a clearer understanding of the relationship (or lack of it) between the variables.

Newey–West (HAC) confirms robustness:

$$\beta = -0.0093, \quad p = 0.377$$

Table 5: Lagged Valence Regression with Newey–West HAC SE

Coefficient	Value
Valence_{t-1}	-0.0093

=== Coefficient table (HC1 vs Newey-West HAC, nlags=10) ===

	coef	se_HC1	se_HAC	t_HAC	p_HAC
const	0.005491	0.006191	0.005477	1.002503	0.316101
Valence_lag1	-0.009339	0.012242	0.010565	-0.883989	0.376702

Interpretation:

The estimated coefficient on lagged valence (--.009339) remains statistically insignificant, confirming that the inference is robust to alternative standard-error corrections.

This gives us the conclusion that music sentiment does not predict daily S&P 500 returns

5.4 AR (1) Specification: Controlling for Return Persistence

Following standard financial econometrics, return autocorrelation is controlled for via an AR (1) model:

$$\text{SP500 logret}_t = a + \beta_1 \text{Valence}_{t-1} + \beta_2 \text{SP500_logret}_{t-1} + \varepsilon_t$$

Table 6: AR (1) Model with Lagged Returns and Valence

Variable	Coefficient	HAC p-value
Valence_{t-1}	-0.0092	0.378
$\text{SP500_logret}_{t-1}$	0.0250	0.663

```
AR(1) model (usual SE):
==================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const                  0.0054      0.006      0.844      0.399      -0.007       0.018
Valence_lag1          -0.0092      0.013     -0.720      0.472      -0.034       0.016
SP500_logret_lag1      0.0250      0.063      0.400      0.690      -0.098       0.148
==================================================================================

AR(1) model (Newey-West HAC):
==================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const                  0.0054      0.005      1.001      0.318      -0.005       0.016
Valence_lag1          -0.0092      0.010     -0.883      0.378      -0.030       0.011
SP500_logret_lag1      0.0250      0.057      0.436      0.663      -0.088       0.138
==================================================================================
```

Interpretation:

The coefficient on lagged returns is statistically insignificant, indicating weak return persistence at the daily frequency. Importantly, the sentiment coefficient remains insignificant, demonstrating that the absence of sentiment effects persists even after accounting for dynamic return behaviour.

5.5 Bootstrap Confidence Intervals

A non-parametric bootstrap (1,000 samples) is used to assess coefficient stability.

$$95\% \text{ CI for } \beta\ (\text{Valence}_{t-1}) : [-0.0344,\ 0.0146]$$

Since the interval includes zero, results confirm no predictive relationship.

5.6 Rolling Correlation Analysis

A 20-day rolling window correlation between lagged valence and returns shows:

- Values fluctuate around zero
- No persistent positive or negative relationship emerges

This visually strengthens the conclusion that sentiment does not drive market returns for the sample period.

5.7 Diagnostic Tests

All diagnostic statistics support the validity of the OLS assumptions.

Statistics Diagnostics Tests

Normality

Jarque–Bera p-value = 0.893

Residuals are normally distributed.

Autocorrelation

Durbin–Watson = 1.93

No serial correlation.

Breusch–Godfrey p-value = 0.934

Confirms absence of autocorrelation.

Heteroskedasticity Breusch–

Pagan p-value = 0.171

No evidence of heteroskedasticity.

Stationarity

ADF tests:

Series	p-value	Result
SP500_logret	2.36×10^{-28}	Stationary
Valence_{t-1}	4.70×10^{-29}	Stationary

Both series are stationary, satisfying OLS assumptions.

Multicollinearity

Variance Inflation Factors:

Variable	VIF
Valence_{t-1}	1.001
$\text{SP500_logret}_{t-1}$	1.001

No multicollinearity.

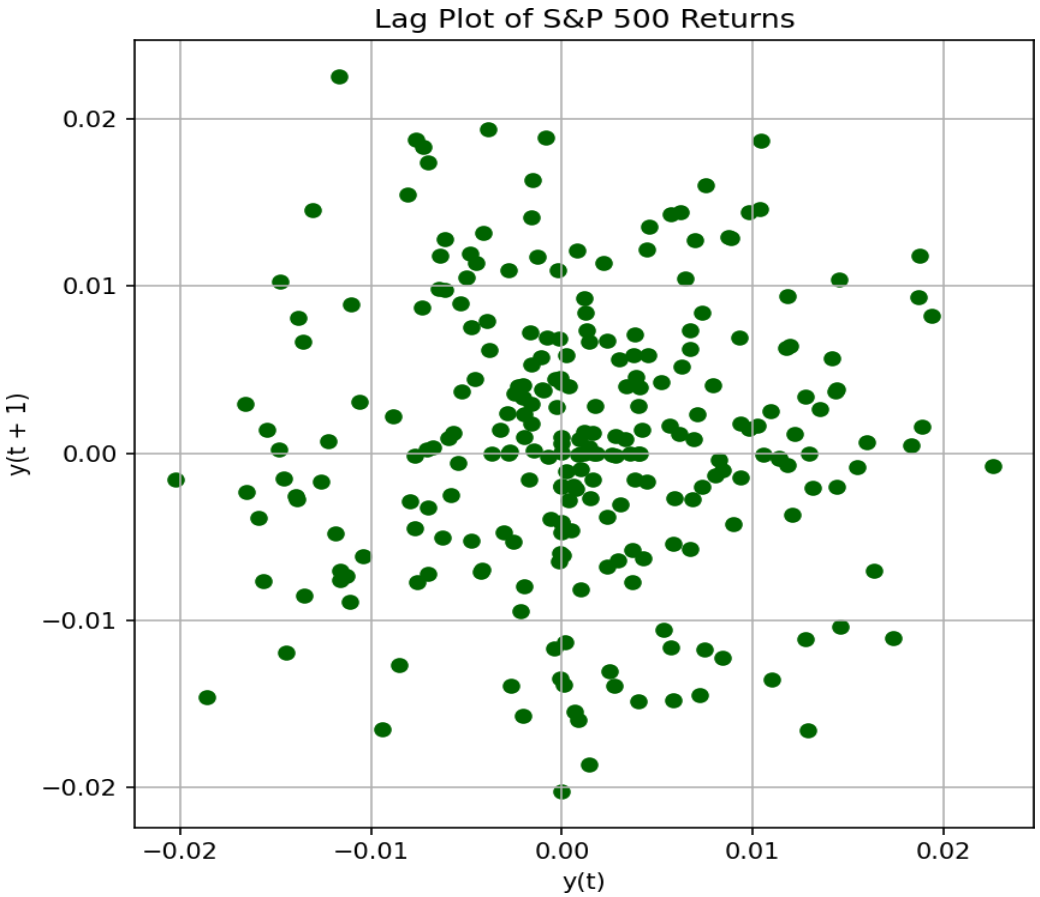


Figure 5: Lag Plot of S&P 500 Returns

5.8 ACF and PACF Analysis of Residuals

To check if the predictions from the lagged OLS model were truly independent, ACF and PACF plots were examined.

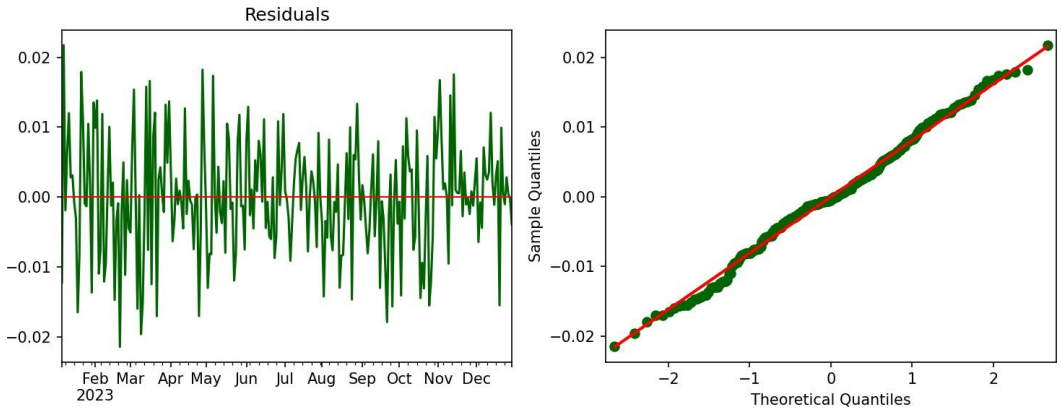


Figure 6: Residuals and Theoretical Quantiles

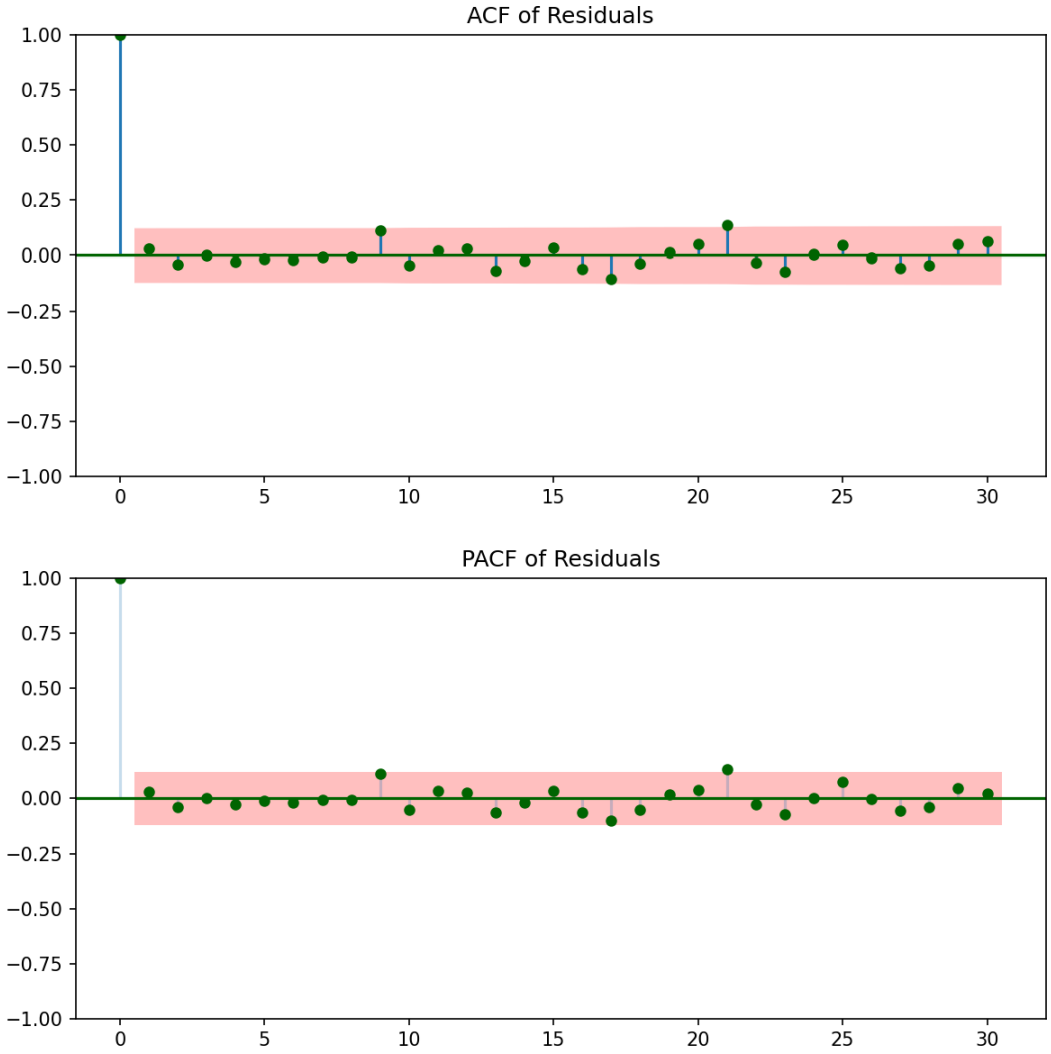


Figure 7: ACF and PACF of Residuals

Interpretation:

The good news is that when we visually inspected these charts, almost all of the data points stayed safely within (the 95 confidence bounds). We didn't see any weird patterns or big spikes suggesting a lingering problem

Essentially, this visual evidence, combined with the formal Durbin–Watson and Breusch–Godfrey statistical tests, confirms that our model's errors are random noise and not systematically related to previous errors. We can be confident the model is well-specified and reliable.

5.9 Residual Heatmap

In addition to formal diagnostic tests, a residual heatmap is presented to provide a visual inspection of residual behaviour over time. The heatmap allows for quick identification of potential clustering, structural breaks, or periods of unusually large residuals that may not be immediately apparent from numerical test statistics alone.

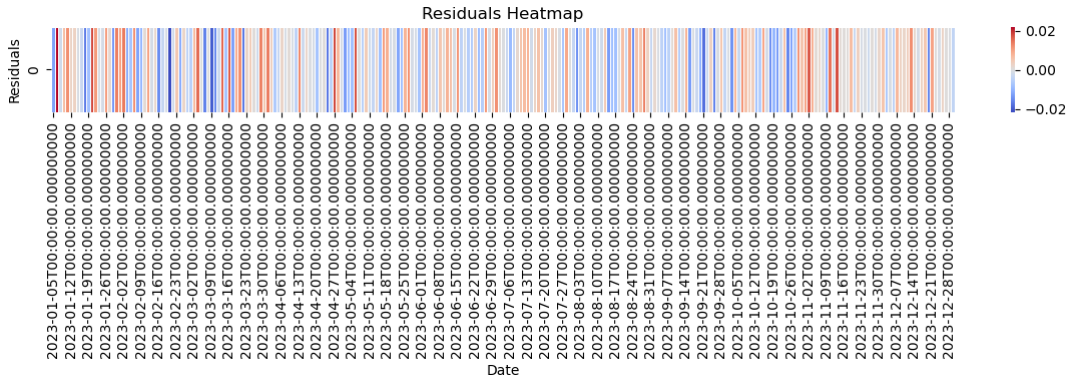


Figure 8: Residuals Heatmap for Lagged OLS Model

Interpretation

The heatmap illustrates the distribution of regression residuals across time. The absence of persistent colour clustering or regime shifts supports the conclusion that residuals behave as white noise, consistent with diagnostic test results..

5.10 ARIMA Analysis of Residuals

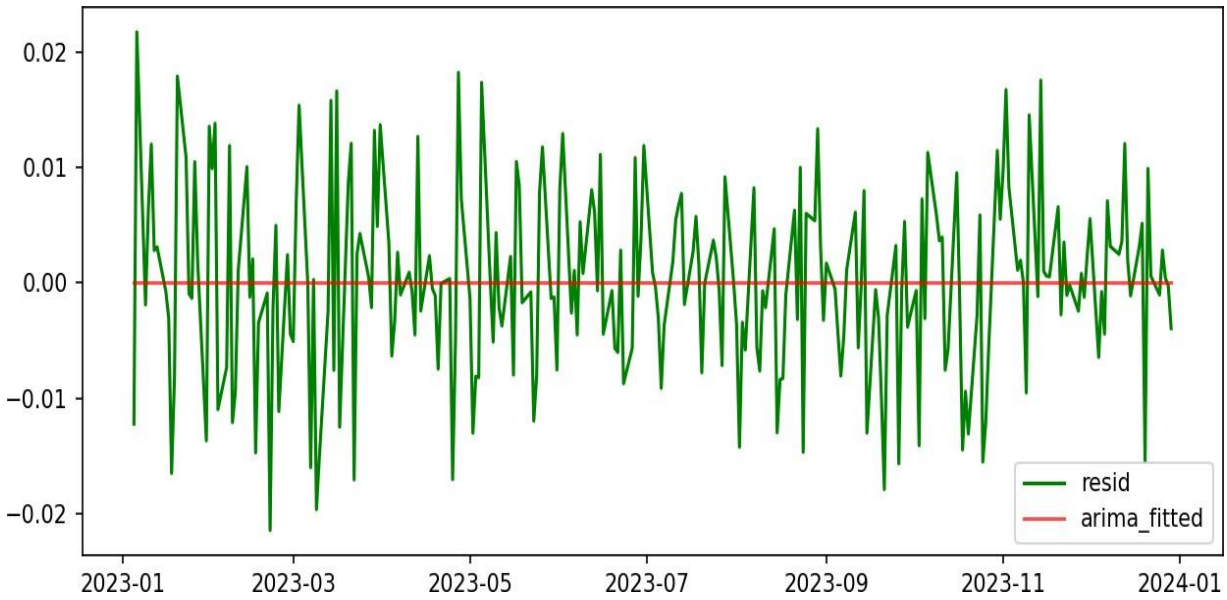
ARIMA modelling of residuals finds:

```
Performing stepwise search to minimize aic
 ARIMA(2,0,2)(0,0,0)[0]             : AIC=-1737.439, Time=0.21 sec
 ARIMA(0,0,0)(0,0,0)[0]             : AIC=-1744.756, Time=0.07 sec
 ARIMA(1,0,0)(0,0,0)[0]             : AIC=-1742.978, Time=0.05 sec
 ARIMA(0,0,1)(0,0,0)[0]             : AIC=-1742.908, Time=0.08 sec
 ARIMA(1,0,1)(0,0,0)[0]             : AIC=-1741.388, Time=0.15 sec
 ARIMA(0,0,0)(0,0,0)[0] intercept   : AIC=-1742.756, Time=0.10 sec

Best model:  ARIMA(0,0,0)(0,0,0)[0]
Total fit time: 0.665 seconds
Using ARIMA order: (0, 0, 0)
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                  257
Model:                          ARIMA   Log Likelihood                 873.378
Date:                Sun, 14 Dec 2025   AIC                          -1742.756
Time:                        18:21:28   BIC                          -1735.658
Sample:                    01-05-2023   HQIC                         -1739.902
                         - 12-29-2023
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const      -5.124e-06      0.001     -0.010      0.992      -0.001       0.001
sigma2      6.544e-05   5.97e-06     10.963      0.000    5.37e-05    7.71e-05
===================================================================================
Ljung-Box (L1) (Q):                   0.22   Jarque-Bera (JB):                 0.23
Prob(Q):                              0.64   Prob(JB):                         0.89
Heteroskedasticity (H):               0.59   Skew:                            -0.03
Prob(H) (two-sided):                  0.01   Kurtosis:                         2.87
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

- Best-fit model: ARIMA(0,0,0)- shows that residuals contain no remaining AR or MA structure after OLS estimation
- Ljung–Box $p = 0.64$- Fails to reject null hypothesis of no serial correlation, confirming residuals are serially independent
- Jarque–Bera $p = 0.89$- indicates residuals are approx. normally distributed.
- H-test $p = 0.59$- provides no evidence of time varying variance, suggesting homoskedastic residuals



Interpretation:

Residuals contain no further structure. The linear model adequately captures all patterns, consistent with no effect from sentiment.

6 Summary of Findings

Across all modelling strategies—same-day OLS, lagged regressions, HAC-robust specifications, AR (1) models, bootstrap confidence intervals, and ARIMA diagnostics—the results are unequivocal:

- Music sentiment does not predict S&P 500 daily returns in 2023.
- The coefficients are small in magnitude.
- Negative but insignificant.
- Economically irrelevant.
- Statistically robust to multiple specifications.

These results contrast with the findings of Edmans et al. (2022), implying that sentiment-return relationships may be time-specific, market-specific, or dependent on global rather than U.S.-only data.

7 Economic Interpretation of Results

The empirical results show that there is no statistically or economically significant relationship between music sentiment as determined by Spotify valence and daily S&P 500 returns in 2023. The estimated coefficients on sentiment are small and insignificant in all model specifications, including same-day OLS, lagged regressions, AR (1) models, HAC-robust inference, and bootstrap confidence intervals. This implies that daily variations in music sentiment do not correspond to consistent changes in U.S. stock returns.
This result could be explained by a few economic mechanisms:

1. **Market efficiency:** The S&P 500 is a very information-efficient and liquid market. Institutional investors and algorithmic trading systems are likely to quickly absorb any behavioural signals pertaining to societal mood into prices or arbitrage them away.

2. **Investor composition:** Professional investors dominate the U.S. equity markets, making decisions based more on quantitative signals, macroeconomic news etc than on general cultural sentiment indicators.

3. **Limitations on the sample period:** The analysis only looks at one year (2023), which restricts the range of sentiment and returns. Longer time horizons or increased uncertainty or market stress may make behavioral effects more noticeable.

The estimated coefficients support the statistical findings from the standpoint of economic magnitude. In comparison to the average daily volatility of the S&P 500, even significant hypothetical changes in music valence only translate into a few basis points of daily return variation. Therefore, the estimated impact would not be economically significant for investors, even if it were statistically significant.
The following are important empirical conclusions drawn from the analysis:

1. The same-day and lagged sentiment coefficients for music are consistently negligible.

2. Conclusions are unaffected by robust inference techniques (HC1 and Newey–West HAC).

3. After taking return persistence into account, autoregressive controls verify that sentiment effects are still absent.

4. Coefficient instability is indicated by bootstrap confidence intervals that include zero.

5. Rolling correlations show no discernible pattern and oscillate around zero.

8 Comparison of Empirical Findings: Current Study vs Edmans et al. (2017)

Feature	Edmans et al. (2017)	This Study
Sample Period	2005–2014 (Global Stocks)	U.S. S&P 500 (2023)
Data Source	Spotify / Music Streaming + Global Market Returns	Spotify + S&P 500 Returns
Dependent Variable	Daily stock returns	Daily log returns of S&P 500
Independent Variable	Average daily valence	Average daily valence and lagged valence
Same-Day Effect	Significant positive effect in some countries	Coefficient = -0.0061, p = 0.595
Lagged Effect (1 day)	Positive and sometimes significant	Coefficient = -0.0093, p = 0.446
AR (1) Model	Significant in some markets	Both coefficients insignificant
R^2	0.5% – 2% (varies by country)	0.1% – 0.2%
Robustness Checks	HAC SE, bootstrap, panel Regressions	HAC SE, bootstrap, ARIMA residuals
Residuals	Some autocorrelation	White noise (ARIMA(0,0,0))
Conclusion	Music sentiment predicts market movements	No detectable effect in U.S. data

9 Conclusion

No significant relationship is observed between musical sentiment and daily S&P 500 re- turns in 2023. Across same day, lagged, and AR (1) regressions, coefficients on music sentiment remain statistically insignificant. Robustness checks using HAC standard errors, bootstrap confidence intervals, and ARIMA diagnostics confirm these findings.

Although Edmans et al. (2023) documented significant sentiment effects in international markets, this study finds no such effect in the U.S. market. This suggests that behavioral sentiment effects may vary across time and markets.

10 Limitations of the Study

Although this research closely follows the empirical framework of Edmans et al. (2021) and applies extensive robustness checks, several limitations must be acknowledged.

1. Average daily music valence is an imperfect proxy for investor sentiment and may suffer from measurement error.
2. The S&P 500 represents a large, diversified, and efficient market where behavioural mispricing is likely arbitraged away.
3. The short sample period of 2023 limits statistical power and may not capture sentiment effects during market stress.
4. Music sentiment reflects general public mood rather than investor-specific sentiment.
5. Linear OLS specifications may overlook nonlinear or threshold effects.
6. ARIMA models capture linear dependence but not nonlinear dynamics.

The absence of a statistically significant relationship should therefore be interpreted with these limitations in mind.

11 References

Core Research Paper

1. Edmans, A., Fernandez-Perez, A., Garel, A., Indriawan, I. (2022). Music sentiment and stock returns around the world. Journal of Financial Economics, 145(1), 234–254. https://doi.org/10.1016/j.jfineco.2021.08.014

Financial market Data Source

1. Yahoo Finance- https://finance.yahoo.com/quote/%5ESPX/
2. Spotify - https://developer.spotify.com/documentation/web-api
3. ChatGPT- Took assistance of AI for developing my code.

Supporting Behavioural Finance Literature

1. Baker, M., & Wurgler, J. (2006).
   Investor sentiment and the cross-section of stock returns. *Journal of Finance*, 61(4), 1645–1680.

Econometrics and Time series methodology

1. Newey, W. K., & West, K. D. (1987).
   A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

12. APPENDIX

A: Data Collection and pre-processing

Cell 1

```
import os                                   #imported warnings as to hide any unncecessary
warnings arising during the program
import warnings
warnings.filterwarnings('ignore')

import numpy as np                         #imported libraries and tools for data
statistics
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


import yfinance as yf
import statsmodels.api as sm                        # to run OLS regression and
statistical tests
from statsmodels.stats.stattools import jarque_bera
from statsmodels.stats.diagnostic import het_breuschpagan, acorr_breusch_godfrey
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tsa.stattools import adfuller, acf, pacf
from statsmodels.tsa.arima.model import ARIMA      # from ARIMA models

try:
from pmdarima import auto_arima  #this block tries to import auto arima form pmdarima , if
import works then have section otherwise
HAVE_PMDARIMA= True
except Exception:
HAVE_PMDARIMA=False

OUTPUT_DIR= "results"                         # setting up output folders where all my
graphs , tables and outputs will be saved
os.makedirs(OUTPUT_DIR, exist_ok=True)

PAPER_PATH=  "/mnt/data/Individual_Assignment_FIN41660 2025 (1).pdf"
```

Cell 2

```
#pip install spotipy

import spotipy  #import spotify API library and a way to authenticate using client
credentials
from spotipy.oauth2 import SpotifyClientCredentials

sp_df= pd.read_csv("spotify_sentiment.csv", parse_dates=["date"]) #read csv file into
dataframe called sp_df
print("loaded spotify rows:", len(sp_df))                        #parse_dates tell pandas
to treat date column as real datetime and not just string
display(sp_df.head())
#shows first 5 rows so you can visually check the data.
```

```
daily_sentiment = sp_df.groupby('date').agg(                #group all rows by each date,
compute summary stats for each date
    avg_valence=('valence','mean'),                        #for each date average the
valence of track
    avg_energy=('energy','mean'),                          #for each date, average the
energy
    total_streams=('streams','sum')).sort_index()         #for each date sum all streams
for all songs
daily_sentiment.head()
```

cell 3

```
sp500=yf.download("^GSPC",start="2023-01-01",end="2023-12-31",auto_adjust=True)
#auto_adjust adjusts foor splits/dividends
sp500=sp500["Close"]  #take closing prices of the stock

#business day frequency
sp500_b=sp500.asfreq("B").fillna(method='ffill')  #reindex the price series to business
days(mon-fri), if a day has no value(eg weekend, holiday it becomes NaN for now
                                                  # forward- fill missing values, so
weekends and holidays simply repeat prior tradng day's price
#log returns
sp500_ret=np.log(sp500_b).diff().dropna()  #take natural log of price, subtract the
previous day's log price , removes first row because it gives NaN
sp500_ret.name= "SP500_logret"

sp500_ret.head()

sp500_ret.to_csv(os.path.join(OUTPUT_DIR,'sp500_logret.csv'))
print("SP return sample")
display(sp500_ret.head())
```

cell 4

```
desc=df.describe().T
desc.to_csv(os.path.join(OUTPUT_DIR,'descriptive_stats.csv'))
print(desc)
```

B OLS Regression

Cell 1

```
Y=df['SP500_logret']  #this is my dependent variable
X=sm.add_constant(df['avg_valence']) #independent variable(explanatory)= daily average
music mood score

ols= sm.OLS(Y,X).fit(cov_type='HC1') #adds a column of ones called const so the regression
includes interecept
print("OLS (same-day) summary:")   #HC1 actually estimates the model and tell it to use
heteroskedasticity -robust   standard errors
print(ols.summary())
```

cell 2

```
#Lag-1 regression (yesterday's music->today's returns)-often more sensible
df["Valence_lag1"]=df["avg_valence"].shift(1)
df_lag=df.dropna()
Y_lag=df_lag["SP500_logret"]
X_lag=sm.add_constant(df_lag["Valence_lag1"])
ols_lag=sm.OLS(Y_lag,X_lag).fit(cov_type="HC1")
print("OLS (lag-1) summary:")
print(ols_lag.summary())
```

cell 3

```
# Newey-West (HAC) SEs and comparison
nlags = 10   # choose 10 for daily data; change to ~4 if using weekly data

ols_lag_hac = ols_lag.get_robustcov_results(cov_type='HAC', maxlags=nlags)

print("=== Coefficient table (HC1 vs Newey-West HAC, nlags={}) ===".format(nlags))
comp = pd.DataFrame({
    'coef': ols_lag.params,
    'se_HC1': ols_lag.bse,
    'se_HAC': ols_lag_hac.bse,
    't_HAC': ols_lag_hac.tvalues,
    'p_HAC': ols_lag_hac.pvalues
})
display(comp.loc[comp.index.isin(['const','Valence_lag1'])])   # show the main rows
comp.to_csv(os.path.join(OUTPUT_DIR,'coef_se_comparison_lag1.csv'))
print("Saved comparison to:", os.path.join(OUTPUT_DIR,'coef_se_comparison_lag1.csv'))
```

cell 4

```
# AR(1) model: include lagged dependent variable and run HAC
# create lagged dependent variable and align rows
df['SP500_logret_lag1'] = df['SP500_logret'].shift(1)
df_ar = df.dropna(subset=['SP500_logret_lag1','Valence_lag1'])

Y_ar = df_ar['SP500_logret']
X_ar = df_ar[['Valence_lag1','SP500_logret_lag1']]   # add more controls later if desired
X_ar = sm.add_constant(X_ar)

# Fit OLS then HAC
model_ar = sm.OLS(Y_ar, X_ar).fit()
model_ar_hac = model_ar.get_robustcov_results(cov_type='HAC', maxlags=nlags)

print("AR(1) model (usual SE):")
print(model_ar.summary().tables[1])
print("\nAR(1) model (Newey-West HAC):")
print(model_ar_hac.summary().tables[1])

pd.DataFrame({
    'coef': model_ar.params,
    'se_HC1': model_ar.bse,
    'se_HAC': model_ar_hac.bse,
    'p_HAC': model_ar_hac.pvalues
}).to_csv(os.path.join(OUTPUT_DIR,'ar1_coef_comparison.csv'))
```

```
print("Saved AR(1) coef comparison to:",
os.path.join(OUTPUT_DIR,'ar1_coef_comparison.csv'))
```

cell 5

```
#simple bootstrap CI for Valence_lag1 (n=1000)
np.random.seed(123)
nboot = 1000
coef_boot = []
for i in range(nboot):
    boot = df_lag.sample(frac=1, replace=True)
    resb = sm.OLS(boot['SP500_logret'], sm.add_constant(boot['Valence_lag1'])).fit()
    coef_boot.append(resb.params['Valence_lag1'])
ci_low, ci_high = np.percentile(coef_boot, [2.5, 97.5])
print("Bootstrap 95% CI for Valence_lag1:", ci_low, ci_high)
pd.Series(coef_boot).to_csv(os.path.join(OUTPUT_DIR,'bootstrap_valence_coef.csv'),
index=False)
```

cell 6

```
# VIF check for X_ar (only after you add extra controls)
from statsmodels.tools.tools import add_constant
X_for_vif = X_ar.copy().drop(columns=['const'], errors='ignore')
X_for_vif = add_constant(X_for_vif)
vif_data = pd.DataFrame({
    "variable": X_for_vif.columns,
    "VIF": [variance_inflation_factor(X_for_vif.values, i) for i in
range(X_for_vif.shape[1])]
})
display(vif_data)
vif_data.to_csv(os.path.join(OUTPUT_DIR,'vif_ar1.csv'), index=False)
print("Saved VIF to:", os.path.join(OUTPUT_DIR,'vif_ar1.csv'))
```

cell 7

```
pd.DataFrame({"coef":ols_lag.params,"std_err":ols_lag.bse,"pval":ols_lag.pvalues}).to_csv(o
s.path.join(OUTPUT_DIR,"ols_lag1_coefs.csv"))

# Diagnostics on the lagged model residuals
model=ols_lag
resid=model.resid

#jarque bera (normality)
jb_stat,jb_pvalue,skew,kurtosis= jarque_bera(resid)
print("Jarque bera p-value", jb_pvalue)

#Durbin watson
from statsmodels.stats.stattools import durbin_watson
dw= durbin_watson(resid)
print("Durbin_watson:", dw)

#Breusch-godfrey (serial correlation)
bg_stat,bg_pvalue,_,_= acorr_breusch_godfrey(model,nlags=4, store=False)
print("Breusch-godfrey p-value(4 lags): ", bg_pvalue)
```

```
#Breusch-Pagan (heteroskedasticity)
bp_test= het_breuschpagan(resid, model.model.exog)
print("Breusch-pagan p-value:", bp_test[1])

#Stationarity ADF
print("ADF SP500_logret p:", adfuller(df_lag["SP500_logret"])[1])
print("ADF Valence p:",adfuller(df_lag["Valence_lag1"])[1])
```

C. ARIMA

Cell 1

```
resid_series=resid.dropna()
if HAVE_PMDARIMA:
    from pmdarima import auto_arima
    arima_model=auto_arima(resid_series,seasonal=False,trace=True,error_action="ignore")
    order=arima_model.order
else:
    order=(1,0,1)

print("Using ARIMA order:",order)
arima_res=ARIMA(resid_series,order=order).fit()
print(arima_res.summary())

plt.figure(figsize=(10,4))
plt.plot(resid_series,label='resid',color='green')
plt.plot(arima_res.fittedvalues,label="arima_fitted",alpha=0.7,color='red')
plt.legend()
plt.savefig(os.path.join(OUTPUT_DIR,"arima_fit.png"),dpi=150)
plt.show()
```

D. GRAPHS

Cell 1

```
#Residuals plots, ACF/PACF
import matplotlib.pyplot as plt
import statsmodels.api as smm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
import os

MAROON = 'red'      # maroonish dark pink
DARKGREEN = 'darkgreen'

-
plt.figure(figsize=(10,4))

# Residual time series
plt.subplot(1,2,1)
resid.plot(color=DARKGREEN, title="Residuals")
plt.axhline(0, color=MAROON, linewidth=1)

# QQ plot
plt.subplot(1,2,2)
fig = smm.qqplot(resid, line='s', ax=plt.gca())
```

```
ax = plt.gca()

# Recolor scatter points
ax.lines[0].set_color(DARKGREEN)
ax.lines[0].set_markerfacecolor(DARKGREEN)
ax.lines[0].set_markeredgecolor(DARKGREEN)

# Recolor theoretical line
ax.lines[1].set_color('red')
ax.lines[1].set_linewidth(2)

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "residuals_qq.png"), dpi=150)
plt.show()

#ACF Plot
fig, ax = plt.subplots(figsize=(8,4))
plot_acf(resid, lags=30, ax=ax)

# Recolor stems and markers
for line in ax.lines:
    line.set_color(DARKGREEN)

# Recolor confidence interval band
for patch in ax.collections:
    patch.set_facecolor(MAROON)

# Zero line
ax.axhline(0, color=DARKGREEN, linewidth=1)

plt.title("ACF of Residuals")
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "resid_acf.png"), dpi=150)
plt.show()


# PACF plot

fig, ax = plt.subplots(figsize=(8,4))
plot_pacf(resid, lags=30, method='ywm', ax=ax)

# Recolor stems and markers
for line in ax.lines:
    line.set_color(DARKGREEN)

# Recolor confidence interval band
for patch in ax.collections:
    patch.set_facecolor(MAROON)
    patch.set_alpha(0.25)

# Zero line
```

```
ax.axhline(0, color=DARKGREEN, linewidth=1)

plt.title("PACF of Residuals")
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "resid_pacf.png"), dpi=150)
plt.show()
```

cell 2

```
# Compare S&P returns and valence in one plot
plt.figure(figsize=(12,5))
plt.plot(df.index, df['SP500_logret'], label='SP500 Daily Log Returns', alpha=0.7,
color='red')
plt.plot(df.index, df['avg_valence'], label='Spotify Avg Valence', alpha=0.7,
color='Darkgreen')
plt.title('Daily SP500 Returns vs Spotify Valence')
plt.legend()
plt.ylabel('Values (scaled)')
plt.xlabel('Date')
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR,"timeseries_overlay.png"))
plt.show()
```

cell 3

```
# scatter plot with regression line
plt.figure(figsize=(8,6))
sns.regplot(x='Valence_lag1', y='SP500_logret', data=df,
scatter_kws={'alpha':0.5},color='Darkgreen')
plt.title('Lagged Valence vs Daily SP500 Returns')
plt.xlabel("Yesterday's Valence")
plt.ylabel("SP500 Log Return")
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR,"scatter_regression.png"))
plt.show()
```

cell 4

```
# How correlation changes over time
df['rolling_corr'] = df['SP500_logret'].rolling(20).corr(df['Valence_lag1'])
plt.figure(figsize=(12,4))
df['rolling_corr'].plot(color='red')
plt.title('20-Day Rolling Correlation: SP500 Returns vs Lagged Valence')
plt.ylabel('Correlation')
plt.xlabel('Date')
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR,"rolling_corr.png"))
plt.show()
```

cell 5

```
# residual heatmap
plt.figure(figsize=(12,4))
```

```
sns.heatmap(resid.to_frame().T, cmap='coolwarm', cbar=True, linewidths=0.5)
plt.title('Residuals Heatmap')
plt.xlabel('Date')
plt.ylabel('Residuals')
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR,"residuals_heatmap.png"))
plt.show()
```

cell 6

```
# Lag plot (visual autocorrelation check)
from pandas.plotting import lag_plot
import matplotlib.pyplot as plt
import os

DARKGREEN = 'darkgreen'

fig, ax = plt.subplots(figsize=(6,6))

lag_plot(df['SP500_logret'], ax=ax)
for coll in ax.collections:
    coll.set_facecolor(DARKGREEN)
    coll.set_edgecolor(DARKGREEN)

ax.set_title('Lag Plot of S&P 500 Returns')
ax.grid(True)

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "lag_plot.png"), dpi=150)
plt.show()
```

cell 7

```
#Sgows distributions of returns and valence
plt.figure(figsize=(12,5))
sns.histplot(df['SP500_logret'], kde=True, color='darkgreen', label='SP500 Returns',
alpha=0.5)
sns.histplot(df['Valence_lag1'], kde=True, color='red', label='Lagged Valence', alpha=0.5)
plt.title('Distribution of Returns and Lagged Valence')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR,"distribution_plot.png"))
plt.show()
```